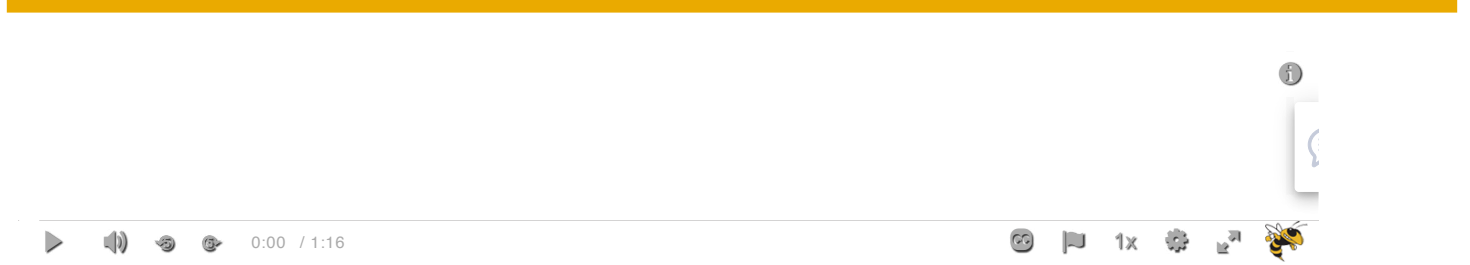


L7: Greedy Modularity Maximization



Computational Complexity

Since the calculation of each ΔM can be done in constant time, Step-2 of the greedy algorithm requires $O(L)$ computations. After deciding which communities to merge, the update of the matrix can be done in a worst-case time $O(N)$. Since the algorithm requires $N-1$ community mergers, its complexity is $O[(L + N)N]$, or $O(N^2)$ on a sparse graph.

Optimized Greedy Algorithm

The use of data structures for sparse matrices can decrease the greedy algorithm's computational complexity to $O(N \log^2 N)$.

For more details please read the paper [Finding community structure in very large networks](http://cs.unm.edu/~aaron/research/fastmodularity.htm)

[by Clauset, Newman and Moore.](http://cs.unm.edu/~aaron/research/fastmodularity.htm)